Numerical Computation — Test 2

Reconstructed Questions

Note. I reconstructed these questions from memory, and the wording was prepared with the help of generative AI (GenAI). This is not an official version of the test. The wording may differ from the original, and some numbers, details or parts of the questions might be incorrect or missing. I also cannot remember all the theory questions from the second test, so some of them might be missing.

1. Newton interpolation

Let $f(x) = \sqrt{x}$. Take the knots

$$x_0 = \frac{1}{4}, \qquad x_1 = 1, \qquad x_2 = 4,$$

with values $f_i = f(x_i)$.

(i) Compute the divided differences $f[x_i]$, $f[x_0, x_1]$, $f[x_1, x_2]$ and $f[x_0, x_1, x_2]$. Use them to produce a quadratic interpolation polynomial in Newton form.

(ii) Add an additional knot $x_3 = 9$. Compute the interpolation polynomial for the knots $x_0, \ldots, x_3$.

(iii) What is an advantage of Newton interpolation over Lagrange interpolation?

2. Basic quadrature rules

Approximate the integral

$$\int_0^1 (2x^3 - 1)\, dx$$

using the midpoint rule, the trapezoidal rule and Simpson's rule, respectively.

Hint: The quadrature point for the midpoint rule is $\frac{1}{2}$. The quadrature points for Simpson's rule are 0, $\frac{1}{2}$ and 1.

3. Degree of exactness

To approximate the integral

$$\int_{-1}^1 f(x)\, dx,$$

consider the quadrature rule

$$Q(f) := f\left(-\frac{1}{3}\right) + f\left(\frac{1}{3}\right).$$

(i) Take $f(x) = x^2 + 2$ and compute the quadrature value $Q(f)$. What is the error compared to the exact integral?

(ii) What is the maximal degree of exactness of this quadrature formula?

4. Relative condition number

The relative condition number for an operation $f : \mathbb{R}^n \to \mathbb{R}$ is the smallest number $\kappa_{\text{rel}}(x)$ such that for all sufficiently small Δx,

$$\frac{|f(x) - f(x + \Delta x)|}{|f(x)|} \leq \kappa_{\text{rel}}(x) \frac{\|\Delta x\|}{\|x\|}.$$

Consider the operation

$$f(x_1, x_2) := 2x_1 + x_2$$

for $x_1, x_2 \geq 0$. Using the definition above, argue whether this operation is well-conditioned with respect to the norm $\|\cdot\| := \|\cdot\|_1$.

5. Cubic splines

Check whether the following function s is a natural, classical cubic spline on $[0, 2]$ with respect to the knots $x_0 = 0$, $x_1 = 1$ and $x_2 = 2$:

$$s(x) := \begin{cases} x^3 - x, & 0 \leq x < 1, \\ x^2 - 1, & 1 \leq x \leq 2. \end{cases}$$

6. Computational complexity

Let $n = 2^p$ for some $p \in \mathbb{N}$. Assume the computational cost $A(n)$ of an algorithm satisfies the recursion

$$A(n) = 2A(n/2), \qquad A(1) = 1.$$

Derive an explicit formula for $A(n)$ and decide whether the computational cost is of order $O(1)$, $O(\sqrt{n})$, $O(n)$, $O(n \ln n)$ or $O(n^2)$.

7. LU factorisation

Consider the linear system $Ax = b$ with

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 10 \\ 30 \end{pmatrix}.$$

Compute the LU factorisation of A without pivoting. Check that your matrices L and U are correct, and solve the linear system $Ax = b$.

8. Orthogonality of Householder reflections

Let $v \in \mathbb{R}^n$ with $\|v\| = 1$. The induced Householder reflection is

$$H = \mathrm{Id} - 2vv^\top.$$

Show that H is an orthogonal matrix.

9. True or false?

For each statement, mark whether it is true or false. No explanation is needed.

(i) Polynomial interpolation. Let P_f denote the Lagrange interpolation polynomial of f with respect to the data points

$$(x_0, f(x_0)), \dots, (x_n, f(x_n)), \qquad a := x_0 < \cdots < x_n =: b.$$

(a) Let $f(x) = x^2 + 1$. Then $f[x_0, x_1, x_2, x_3] = 0$.
☐ True ☐ False

(b) For polynomial interpolation, changing one data value $f(x_i)$ affects the interpolant P_f everywhere except at finitely many points.
☐ True ☐ False

(c) The error $\max_{x\in[a,b]} |P_f(x) - f(x)|$ does not depend on the choice of knots.
☐ True ☐ False

(d) The Lagrange and Newton interpolation polynomials are always identical for the same data points.
☐ True ☐ False

(ii) Conditioning.

(a) Taking the square root of a positive real number is ill-conditioned.
☐ True ☐ False

(b) Multiplication and division of two positive real numbers are always well-conditioned.
☐ True ☐ False

(iii) Quadrature.

(a) The Newton–Cotes formulas converge to the exact integral as the number of quadrature points $n \to \infty$.
☐ True ☐ False

(b) A quadrature rule on $[a, b]$ with $\sum_i w_i = b - a$ integrates constant functions on $[a, b]$ exactly.
☐ True ☐ False

(c) The weights in the Gaussian quadrature formula are positive.
☐ True ☐ False

(d) The weights w_i of the Newton–Cotes formulas depend on the interval $[a, b]$.
☐ True ☐ False

10. Short theory questions

Answer the following questions.

(i) What are the given quantities in the polynomial interpolation problem?

(ii) Let the $n + 1$ knots $x_0 < \cdots < x_n$ be given. Write down the formula for the Lagrange polynomials $(\ell_i)_{i=0}^n$ with respect to the points $(x_i)_{i=0}^n$.

(iii) What is the maximal degree of polynomials that can be integrated exactly by an interpolatory quadrature formula with $n + 1$ knots? Which interpolatory quadrature formula achieves this?

(iv) What is row pivoting in Gaussian elimination? Name two reasons why it is done.

(v) Define the condition number of a matrix $A \in \mathbb{R}^{n\times n}$ with respect to the matrix norm $\|\cdot\|$.